STATISTICAL ANALYSIS & DATA SCIENCE

CO5-AT1 Pseudocode Design

PSEUDOCODE-BASED SOLUTIONS

Assessment Coverage	Questions 16–20
Core Topics	K-Means Clustering, Elbow Method, Classification Metrics, Model Selection
Answer	Structured, and implementation-oriented
Programming Form	Language-independent pseudocode

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16. K-Means Clustering — Basic Algorithm

Question: Design pseudocode to implement the K-Means clustering algorithm: initialize centroids, assign observations to the nearest centroid, update centroids, and repeat until convergence.

Pseudocode

```
BEGIN
INPUT dataset X with n observations and number of clusters K
Choose K initial centroids C[1], C[2], ..., C[K]
REPEAT
    FOR each observation x[i] in X
        FOR each centroid C[j]
            distance[j] ← EuclideanDistance(x[i], C[j])
        END FOR
        cluster[i] ← index of centroid with minimum distance
    END FOR

    FOR each cluster j = 1 to K
        IF cluster j contains observations THEN
            C_new[j] ← mean of all observations assigned to cluster j
        ELSE
```

```
            Reinitialize C_new[j] using an unassigned/random observation
        END IF
    END FOR

    Calculate centroid movement or compare old and new cluster assignments
    C ← C_new
UNTIL centroid movement < tolerance OR assignments do not change OR max iterations
reached

OUTPUT final centroids and cluster assignments
END
```

Explanation

K-Means alternates between two main operations: assigning every observation to its nearest centroid and recomputing each centroid as the mean of the observations assigned to it. The process converges when the centroids or assignments stop changing significantly.

Convergence Condition

A practical stopping rule is: stop when the maximum centroid movement is less than a predefined tolerance, such as 0.001, or when the maximum number of iterations is reached.

17. Customer Segmentation into Three Clusters

Question: Design pseudocode to group customers into three clusters based on annual income and spending score using K-Means clustering.

Pseudocode

```
BEGIN
INPUT customer dataset
Select features: AnnualIncome and SpendingScore
Set K ← 3

Preprocess the two features
    Handle missing values if required
    Scale/standardize AnnualIncome and SpendingScore when their scales differ

Initialize 3 centroids randomly or using K-Means++

REPEAT
    FOR each customer i
        d1 ← distance(customer[i], centroid1)
        d2 ← distance(customer[i], centroid2)
        d3 ← distance(customer[i], centroid3)
        Assign customer[i] to the nearest centroid
    END FOR
```

```
    FOR cluster j = 1 to 3
        centroid[j] ← mean(AnnualIncome, SpendingScore of customers in cluster j)
    END FOR
UNTIL assignments no longer change OR convergence tolerance is reached

OUTPUT cluster label for every customer
OUTPUT final centroids
Interpret clusters using their income and spending-score patterns
END
```

Interpretation

```
Cluster 1 may represent customers with lower income and lower spending.
Cluster 2 may represent customers with higher income and higher spending.
Cluster 3 may represent another distinct segment, such as high-income but low-
spending or moderate-income high-spending customers.
```

Conclusion

The final cluster meaning should be determined from the actual centroid values rather than assuming a fixed customer type before running the algorithm.

18. Selecting the Number of Clusters Using the Elbow Method

Question: Design pseudocode to determine the suitable number of clusters using the Elbow Method and then apply K-Means with the selected number of clusters.

Pseudocode

```
BEGIN
INPUT dataset X
Set K_min ← 2
Set K_max ← 10
Create empty list WCSS

FOR K from K_min to K_max
    Run K-Means on X using K clusters
    Calculate WCSS[K] ← sum of squared distances
        between each observation and its assigned centroid
    Store WCSS[K]
END FOR

Plot K against WCSS
Identify the elbow point where WCSS decreases sharply before
the improvement becomes relatively small
selected_K ← elbow point

Run K-Means again using selected_K
Assign observations to nearest centroid
Update centroids until convergence
```

```
OUTPUT selected_K, final centroids, and cluster labels
END
```

Elbow Criterion

The suitable K is usually the point where adding another cluster gives only a relatively small reduction in WCSS. The method balances compact clusters against unnecessary model complexity.

Decision

For example, if WCSS decreases strongly from K=2 to K=3 and then decreases only gradually from K=3 onward, K=3 can be selected as the elbow point.

19. Evaluation Metrics — TP, TN, FP, FN

Question: Design pseudocode to calculate TP, TN, FP, and FN from actual and predicted binary class labels and subsequently calculate accuracy, precision, recall, and F1-score.

Pseudocode

```
BEGIN
INPUT actual labels Y_true and predicted labels Y_pred
Set TP ← 0, TN ← 0, FP ← 0, FN ← 0

FOR i from 1 to number of observations
    IF Y_true[i] = 1 AND Y_pred[i] = 1 THEN
        TP ← TP + 1
    ELSE IF Y_true[i] = 0 AND Y_pred[i] = 0 THEN
        TN ← TN + 1
    ELSE IF Y_true[i] = 0 AND Y_pred[i] = 1 THEN
        FP ← FP + 1
    ELSE IF Y_true[i] = 1 AND Y_pred[i] = 0 THEN
        FN ← FN + 1
    END IF
END FOR

Accuracy  ← (TP + TN) / (TP + TN + FP + FN)
Precision ← TP / (TP + FP), if TP + FP > 0
Recall    ← TP / (TP + FN), if TP + FN > 0
F1-score  ← 2 × Precision × Recall / (Precision + Recall), if denominator > 0

OUTPUT TP, TN, FP, FN
OUTPUT Accuracy, Precision, Recall, F1-score
END
```

Metric Meaning

```
Accuracy measures the proportion of all predictions that are correct.
Precision measures how many predicted positives are actually positive.
Recall measures how many actual positives are correctly detected.
```

```
F1-score is the harmonic mean of precision and recall and is useful when both are
important.
```

Important Note

When a denominator is zero, the corresponding metric should be handled using a defined policy, such as reporting it as 0 or marking it as undefined.

20. Comparing kNN, Decision Tree, Logistic Regression, and K-Means

Question: Design pseudocode for comparing kNN, Decision Tree, Logistic Regression, and K-Means models using appropriate evaluation measures and selecting the most suitable model for a given Data Science problem.

Pseudocode

```
BEGIN
INPUT dataset D and identify the target variable

IF target is categorical and labels are available THEN
    ProblemType ← Supervised Classification
ELSE IF target is continuous THEN
    ProblemType ← Supervised Regression
ELSE
    ProblemType ← Unsupervised Clustering
END IF

Preprocess data
    Handle missing values
    Encode categorical features
    Scale features when appropriate, especially for kNN
    Split labeled data into training and test sets when supervised

IF classification THEN
    Train kNN
    Train Decision Tree
    Train Logistic Regression
    Evaluate each using accuracy, precision, recall, F1-score
    Optionally use cross-validation for reliable comparison
    Select the model with the best validation performance
    considering the application's priority and computational constraints

ELSE IF clustering THEN
    Determine suitable K using Elbow Method or another clustering criterion
    Train K-Means with selected K
    Evaluate using Silhouette Score, WCSS, and cluster interpretability
    Select K-Means only when clustering is appropriate for the problem
```

```
ELSE IF regression THEN
    Do not use classification-only metrics
    Select an appropriate regression model instead
END IF

Compare performance, interpretability, speed, robustness, and deployment needs
OUTPUT selected model and justification
END
```

Evaluation Framework

```
kNN: For classification, use accuracy, precision, recall, F1-score, and cross-
validation. Scaling is important because kNN relies on distances.
Decision Tree: Use classification metrics for classification tasks. Consider
interpretability, overfitting, tree depth, and validation performance.
Logistic Regression: Use accuracy, precision, recall, F1-score, ROC-AUC where
appropriate, and probability-based analysis.
K-Means: It is unsupervised, so class-label metrics such as accuracy are generally
inappropriate. Use WCSS/inertia, Silhouette Score, cluster separation, stability, and
business interpretability.
```

Model Selection

There is no universally best model. For a labeled binary or multiclass classification problem, compare kNN, Decision Tree, and Logistic Regression using the same validation procedure and select the model that best satisfies the required metric and deployment constraints. If the task is customer segmentation without target labels, K-Means is more appropriate because the objective is to discover natural groups rather than predict a known class.

Conclusion

The final selection should be based on the problem type first, followed by evaluation performance, interpretability, computational cost, scalability, and real-world requirements. This prevents choosing a model solely because it has a high score on an inappropriate metric.

Overall Conclusion

Questions 16–20 demonstrate the complete workflow from clustering implementation to model evaluation and selection. K-Means repeatedly performs assignment and centroid-update steps until convergence. The Elbow Method helps select a practical number of clusters, while classification metrics provide a structured way to evaluate binary predictions. Finally, model selection must consider both quantitative performance and the nature of the Data Science problem. In particular, K-Means should be evaluated with clustering measures rather than ordinary classification accuracy.

Key Takeaways

- K-Means assigns observations to the nearest centroid and updates centroids using cluster means.
- Customer segmentation can use Annual Income and Spending Score as clustering features.
- The Elbow Method uses WCSS to identify a practical value of K.

- TP, TN, FP, and FN form the basis of accuracy, precision, recall, and F1-score.
- Model evaluation measures must match the learning task: supervised classification and unsupervised clustering require different criteria.